Lab - 1 | 22 January 2026

↳ 6 cells hidden

Lab - 2 | 29 January 2026

↳ 14 cells hidden

LAB -3 5 Feb

↳ 2 cells hidden

Lab- 4 | 12 Feb

↳ 20 cells hidden

Lab - 5 | 19 Feb

↳ 1 cell hidden

Lab 6 || 26 Feb

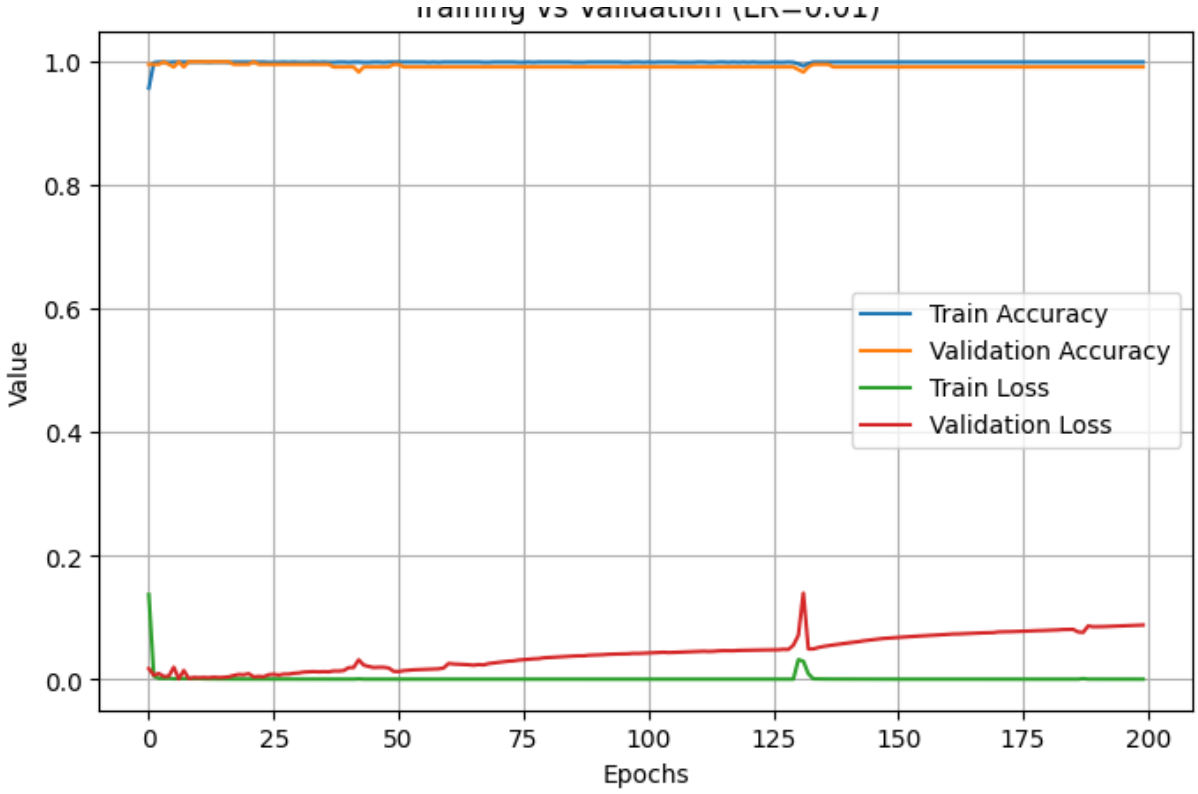
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_
from sklearn.feature_extraction.text import TfidfVectorizer

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

```
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam


df = pd.read_csv('/spam_or_not_spam.csv')
X_text = df['email'].fillna('')
y = df['label']


vectorizer = TfidfVectorizer(max_features=5000)
X = vectorizer.fit_transform(X_text).toarray()


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ================================
# 4. Build Model
# ================================
def create_model(lr):
    model = Sequential()
    model.add(Dense(64, activation='relu', input_shape=(X.shape[1],)
    model.add(Dense(32, activation='relu'))
    model.add(Dense(1, activation='sigmoid'))

    model.compile(
        optimizer=Adam(learning_rate=lr),
        loss='binary_crossentropy',
        metrics=['accuracy']
    )
    return model


# ================================
# 5. Training + Graph
# ================================
learning_rates = [0.01, 0.001, 0.005]

for lr in learning_rates:

    print("\n==============================")
    print("Training with LR =", lr)

    model = create_model(lr)

    history = model.fit(
        X_train, y_train,
        epochs=200,
```

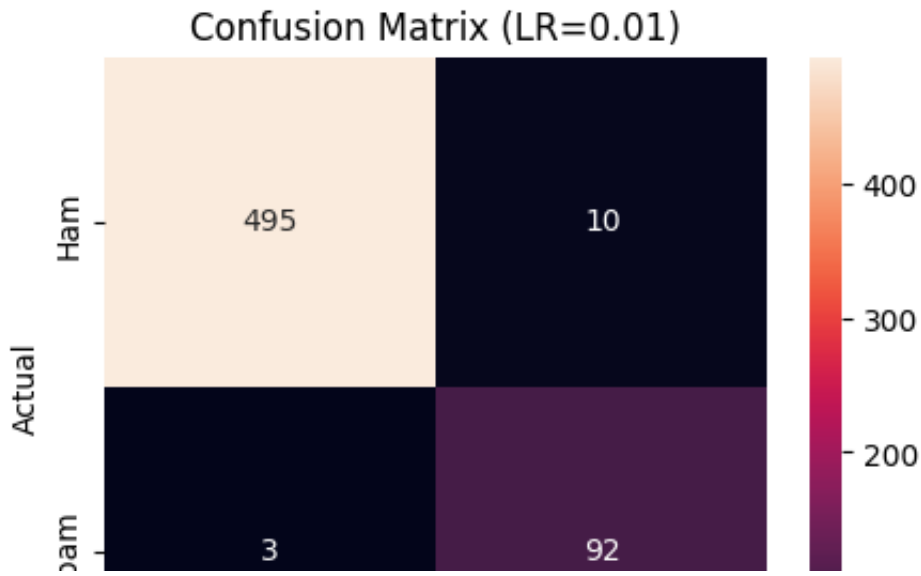
```
            batch_size=32,
            validation_split=0.1,
            verbose=0
        )

        plt.figure(figsize=(8,5))
        plt.plot(history.history['accuracy'], label='Train Accuracy')
        plt.plot(history.history['val_accuracy'], label='Validation Accu
        plt.plot(history.history['loss'], label='Train Loss')
        plt.plot(history.history['val_loss'], label='Validation Loss')
        plt.title(f"Training vs Validation (LR={lr})")
        plt.xlabel("Epochs")
        plt.ylabel("Value")
        plt.legend()
        plt.grid(True)
        plt.show()


        y_pred = (model.predict(X_test) > 0.5).astype("int32")

        # ================================
        # Metrics
        # ================================
        print("Accuracy :", accuracy_score(y_test, y_pred))
        print("Precision:", precision_score(y_test, y_pred))
        print("Recall   :", recall_score(y_test, y_pred))
        print("F1 Score :", f1_score(y_test, y_pred))
        print("\nClassification Report:\n", classification_report(y_test

        # ================================
        # Confusion Matrix Heatmap
        # ================================
        cm = confusion_matrix(y_test, y_pred);

        plt.figure(figsize=(5,4))
        sns.heatmap(cm, annot=True, fmt='d',
                    xticklabels=['Ham', 'Spam'],
                    yticklabels=['Ham', 'Spam'])
        plt.title(f"Confusion Matrix (LR={lr})")
        plt.xlabel("Predicted")
        plt.ylabel("Actual")
        plt.show()
```

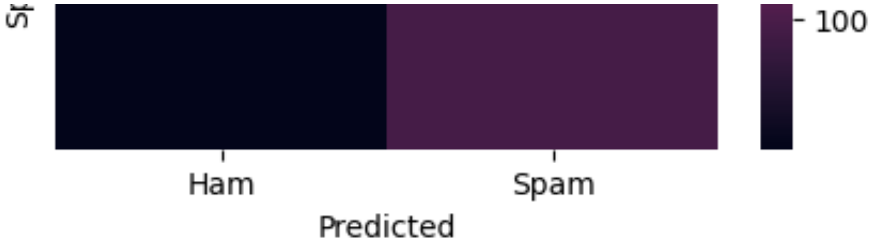
```
================================
Training with LR = 0.01
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.
  super().__init__(activity_regularizer=activity_regularizer, **kwar
```

Training vs Validation (LR=0.01)

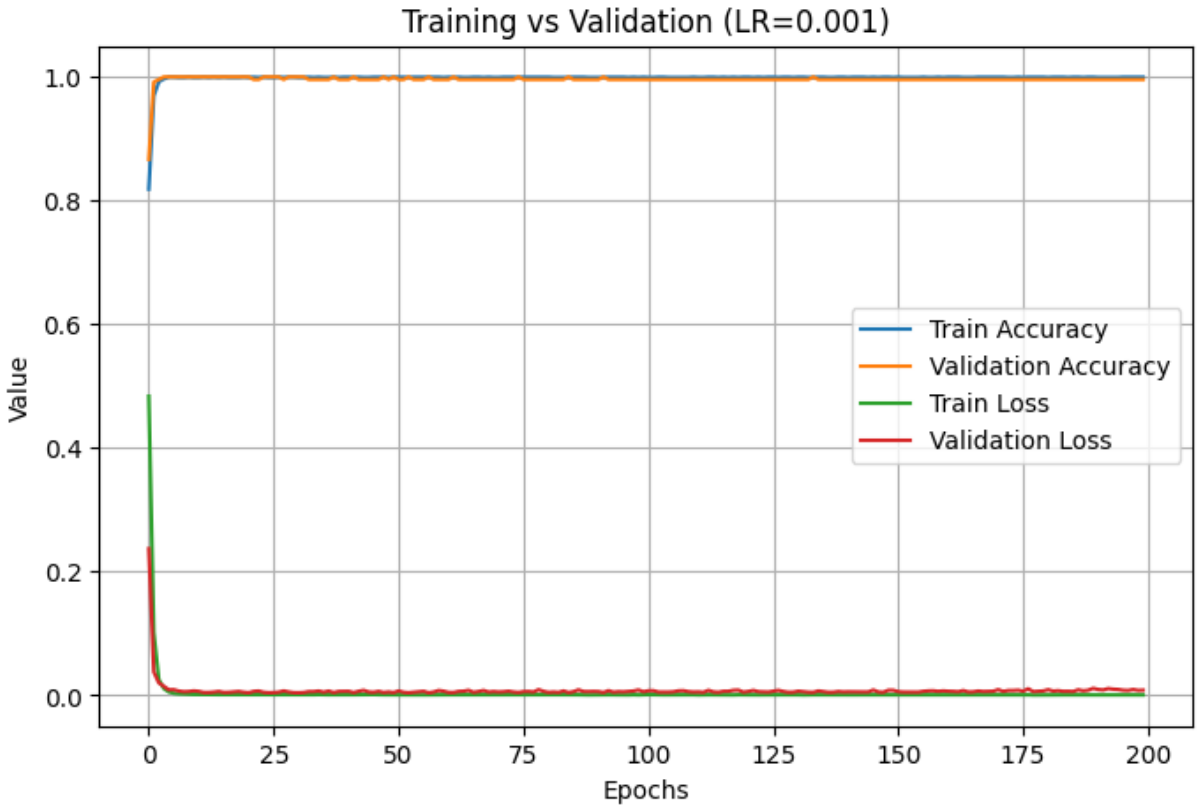



```
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Accuracy : 0.9783333333333334
Precision: 0.9019607843137255
Recall   : 0.968421052631579
F1 Score : 0.934010152284264

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.98      0.99       505
           1       0.90      0.97      0.93        95

    accuracy                           0.98       600
   macro avg       0.95      0.97      0.96       600
weighted avg       0.98      0.98      0.98       600
```

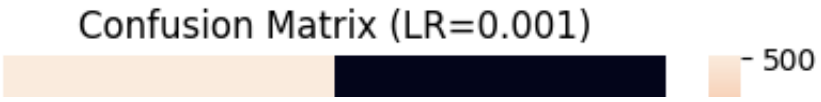


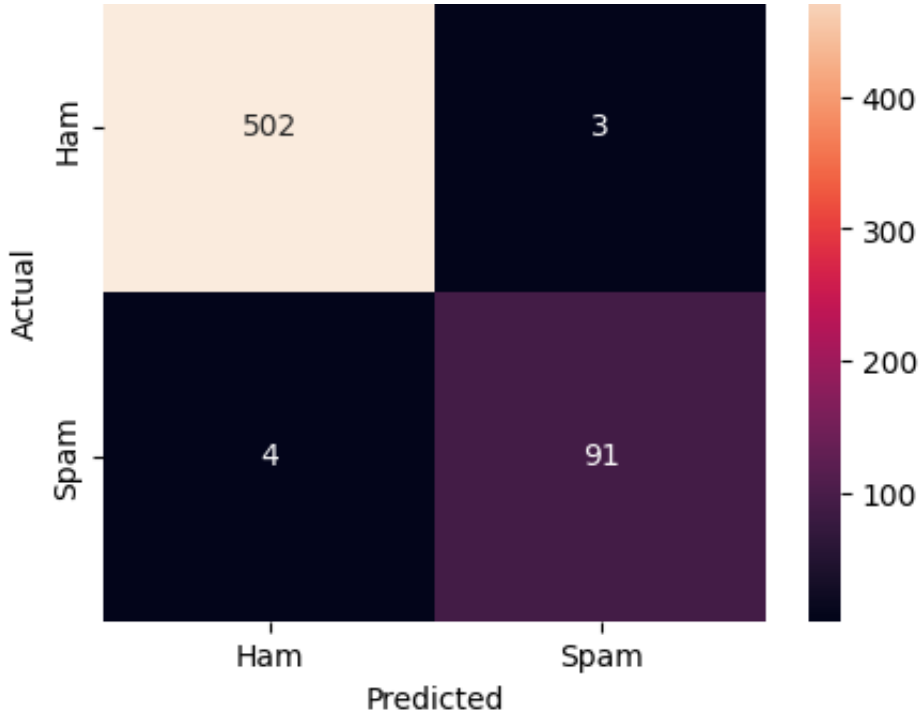



```
==============================
Training with LR = 0.001
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.
  super().__init__(activity_regularizer=activity_regularizer, **kwar
```

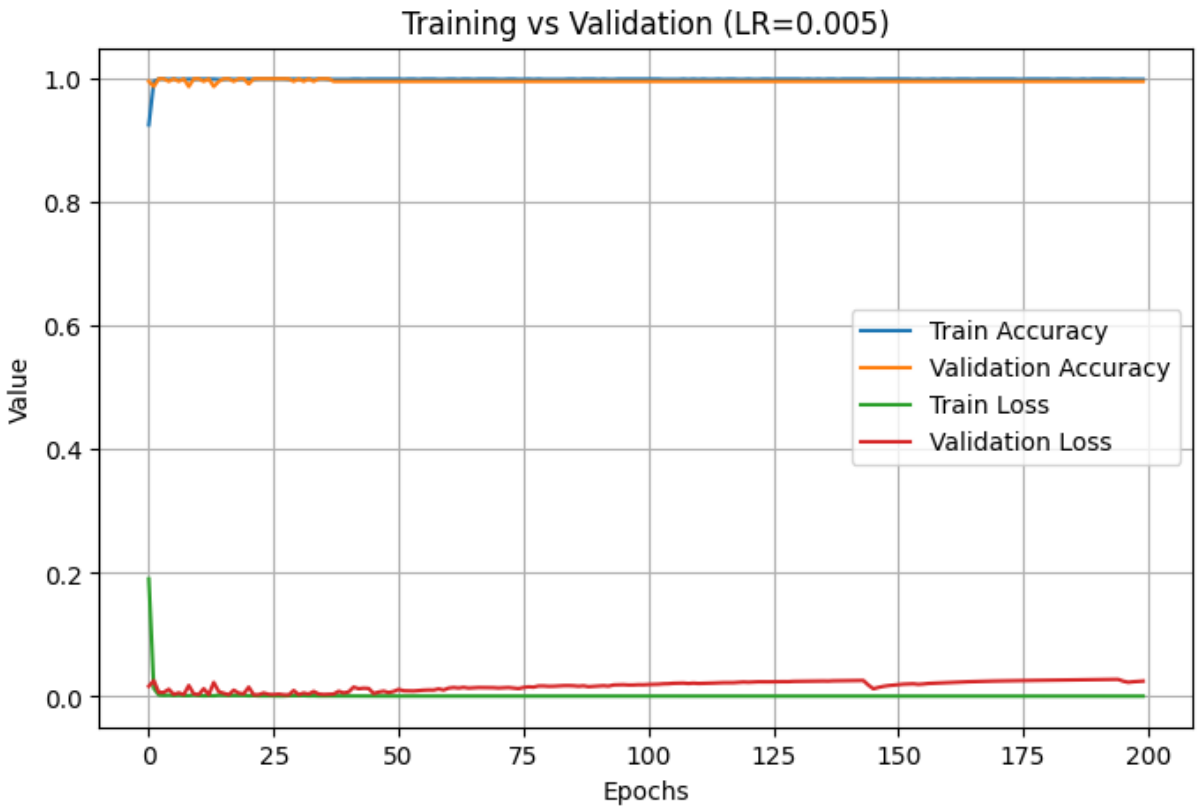



```
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Accuracy : 0.9883333333333333
Precision: 0.9680851063829787
Recall   : 0.9578947368421052
F1 Score : 0.9629629629629629

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       505
           1       0.97      0.96      0.96        95

    accuracy                           0.99       600
   macro avg       0.98      0.98      0.98       600
weighted avg       0.99      0.99      0.99       600
```



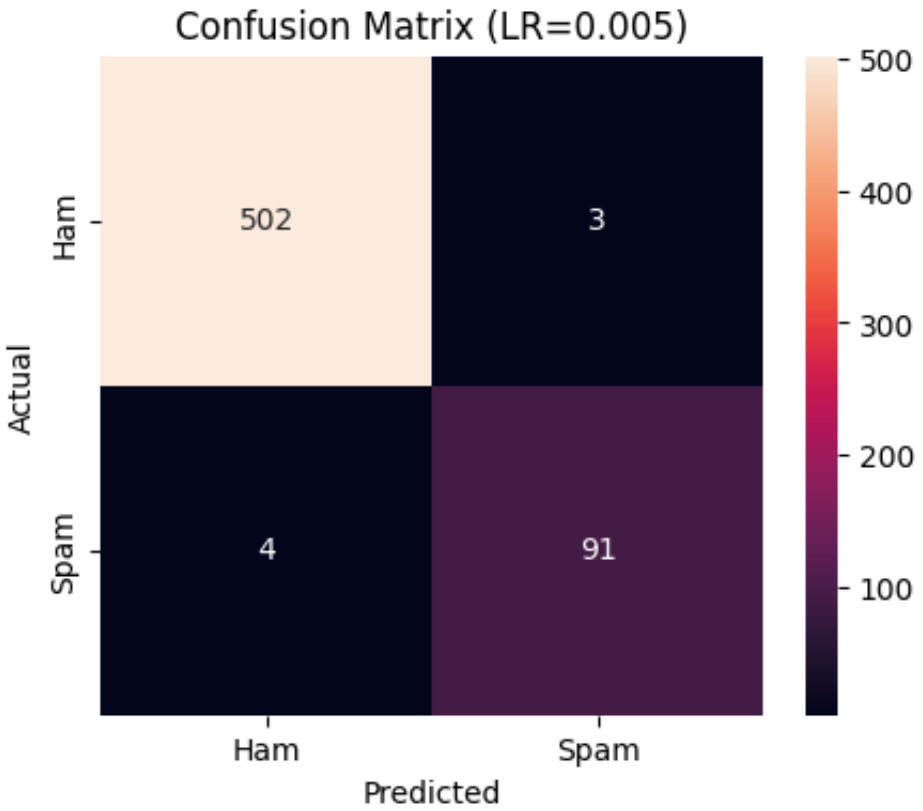



```
==============================
Training with LR = 0.005
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.
  super().__init__(activity_regularizer=activity_regularizer, **kwar
```




```
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Accuracy : 0.9883333333333333
Precision: 0.9680851063829787
Recall   : 0.9578947368421052
F1 Score : 0.9629629629629629
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       505
           1       0.97      0.96      0.96        95

    accuracy                           0.99       600
   macro avg       0.98      0.98      0.98       600
weighted avg       0.99      0.99      0.99       600
```



Start coding or generate with AI.